Comparing two independent builds of the armed conflict analytical data set

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1 What is being compared

`final_data.csv` was built in the week 3 in-class assignment by `R/create_final_data.R`, which uses the tidyverse: `pivot_longer()` for the World Bank files, `pivot_wider()` for the disaster indicators, `group_by()` and `summarise()` for the conflict deaths, and `left_join()` to merge.

`final_data_agent.csv` was built by `R/create_final_data_agent.R`, which a coding agent wrote from the same written specification without reusing any of that code. It uses base R only: `reshape()`, `aggregate()`, `merge()` and `split()`. The point of writing it twice in two different dialects is that an error in one implementation is unlikely to repeat itself in the other, so any disagreement between the two outputs points at something worth checking.

```
own <- read.csv(here("data", "processed", "final_data.csv"))
agent <- read.csv(here("data", "processed", "final_data_agent.csv"))
```

2 Shape and column names

```
tibble(
  Check = c("Rows", "Columns", "Column names identical", "Column order identical"),
  Own = c(nrow(own), ncol(own), "", ""),
  Agent = c(nrow(agent), ncol(agent), "", ""),
  Agree = c(nrow(own) == nrow(agent),
            ncol(own) == ncol(agent),
            setequal(names(own), names(agent)),
            identical(names(own), names(agent)))
```

```
) |>
  kable(caption = "Dimensions and column names")
```

Table 1: Dimensions and column names

Check	Own	Agent	Agree
Rows	3720	3720	TRUE
Columns	18	18	TRUE
Column names identical			TRUE
Column order identical			TRUE

Both files have 3720 rows, which is exactly 186 countries times 20 years (2000 to 2019). That is the balanced panel the paper describes, so neither build dropped or duplicated a country-year.

3 Cell by cell comparison

```
agent <- agent[, names(own)]
own <- own[order(own$iso, own$year), ]
agent <- agent[order(agent$iso, agent$year), ]
row.names(own) <- NULL
row.names(agent) <- NULL

comparison <- all.equal(own, agent)
identical_output <- isTRUE(comparison)
identical_output
```

```
[1] TRUE
```

```
mismatch_count <- function(x, y) {
  both_na <- is.na(x) & is.na(y)
  sum(!both_na & (is.na(x) != is.na(y) | x != y), na.rm = TRUE)
}

map_int(names(own), \(v) mismatch_count(own[[v]], agent[[v]])) |>
  set_names(names(own)) |>
  enframe(name = "Variable", value = "Disagreeing cells") |>
  kable(caption = "Number of country-years where the two builds disagree")
```

Table 2: Number of country-years where the two builds disagree

Variable	Disagreeing cells
iso	0
year	0
matmor	0
infmor	0
neomor	0
un5mor	0
armedconflict	0
earthquake	0
drought	0
gdp_1000	0
oecd	0
pop_dens	0
urban	0
age_dep	0
male_edu	0
temp	0
rainfall_1000	0
armedconflict_lag1	0

Every column agrees in every one of the 3720 rows, missing values included. The two implementations produce the same data set.

4 Checks on the derived variables

Agreement between two implementations only shows that they made the same choices. The checks below ask whether those choices match the paper.

```
tibble(
  Quantity = c("Countries", "Years", "Country-years with armed conflict",
               "Country-years with an earthquake",
               "Country-years with a drought",
               "First year of lagged exposure that is missing"),
  Value = c(n_distinct(own$iso),
            paste(min(own$year), "to", max(own$year)),
            sum(own$armedconflict),
            sum(own$earthquake),
            sum(own$drought),
```

```
            sum(is.na(own$armedconflict_lag1)))
) |>
  kable(caption = "Derived quantities in the merged data")
```

Table 3: Derived quantities in the merged data

Quantity	Value
Countries	186
Years	2000 to 2019
Country-years with armed conflict	665
Country-years with an earthquake	310
Country-years with a drought	325
First year of lagged exposure that is missing	186

The lagged exposure is missing for exactly 186 rows, one per country, which is the year 2000 row where no earlier year exists. The lag was computed within country, so no country inherits the previous country's last value.

```
own |>
  filter(iso %in% c("AFG", "CAN")) |>
  group_by(iso) |>
  slice_head(n = 3) |>
  select(iso, year, armedconflict, armedconflict_lag1) |>
  kable(caption = "The lag resets at each country")
```

Table 4: The lag resets at each country

iso	year	armedconflict	armedconflict_lag1
AFG	2000	1	NA
AFG	2001	1	1
AFG	2002	1	1
CAN	2000	0	NA
CAN	2001	0	0
CAN	2002	0	0

5 Where the two builds could have diverged

Three judgment calls in the specification are not fully determined by the assignment text. Both builds resolved them the same way, so they do not show up as disagreements between the two

files, but each of them changes the data if resolved differently, and they are the first places to look if these numbers do not match a classmate's.

5.1 Which year column defines a disaster

```
dis_raw <- read.csv(here("data", "raw", "disaster.csv"), check.names = TRUE) |>
  filter(Disaster.Type %in% c("Earthquake", "Drought"))

tibble(
  Check = c("Earthquake and drought rows in the file",
            "Rows kept by `Year` in 2000 to 2019",
            "Rows kept by `Start Year` in 2000 to 2019",
            "Rows where `Year` and `Start Year` differ"),
  Rows = c(nrow(dis_raw),
           sum(dis_raw$Year >= 2000 & dis_raw$Year <= 2019),
           sum(dis_raw$Start.Year >= 2000 & dis_raw$Start.Year <= 2019),
           sum(dis_raw$Year != dis_raw$Start.Year, na.rm = TRUE))
) |>
  kable(caption = "Disaster year: the `Year` column versus the `Start Year` column")
```

Table 5: Disaster year: the `Year` column versus the `Start Year` column

Check	Rows
Earthquake and drought rows in the file	1544
Rows kept by `Year` in 2000 to 2019	890
Rows kept by `Start Year` in 2000 to 2019	893
Rows where `Year` and `Start Year` differ	25

The two columns are not interchangeable. They disagree on 25 rows, all of them droughts, where EM-DAT records the year the drought was entered under and a later year as the year it began.

```
dis_raw |>
  filter(Year != Start.Year) |>
  count(Disaster.Type, name = "Rows") |>
  kable(caption = "Which disaster types have a `Year` that differs from `Start Year`")
```

Table 6: Which disaster types have a `Year` that differs from `Start Year`

Disaster.Type	Rows
Drought	25

The table below rebuilds the disaster indicators under each choice.

```
build_disaster <- function(year_col) {
  keep <- dis_raw[dis_raw[[year_col]] >= 2000 & dis_raw[[year_col]] <= 2019, ]
  data.frame(iso = keep$ISO,
             year = keep[[year_col]],
             type = keep$Disaster.Type) |>
    distinct() |>
    mutate(value = 1) |>
    pivot_wider(names_from = type, values_from = value, values_fill = 0) |>
    janitor::clean_names()
}

by_year <- build_disaster("Year")
by_start <- build_disaster("Start.Year")

differing <- full_join(by_year, by_start, by = c("iso", "year"),
                       suffix = c("_year", "_start")) |>
  mutate(across(where(is.numeric), \(x) replace_na(x, 0))) |>
  filter(earthquake_year != earthquake_start | drought_year != drought_start)

tibble(
  Quantity = c("Country-years with a disaster",
               "Earthquake country-years",
               "Drought country-years"),
  `Using Year` = c(nrow(by_year), sum(by_year$earthquake), sum(by_year$drought)),
  `Using Start Year` = c(nrow(by_start), sum(by_start$earthquake),
                         sum(by_start$drought))
) |>
  kable(caption = "The disaster indicators under each year definition")
```

Table 7: The disaster indicators under each year definition

Quantity	Using Year	Using Start Year
Country-years with a disaster	600	603

Quantity	Using Year	Using Start Year
Earthquake country-years	320	320
Drought country-years	327	330

```
nrow(differing)
```

```
[1] 21
```

Switching to `Start Year` would move 21 country-years, all of them droughts. Both builds use `Year`, which is what the week 3 instructions named, so the two files agree, but this is a real choice with a real effect and not a difference that can be ignored.

5.2 Repeated disasters within one country-year

```
dis_raw |>
  filter(Year >= 2000, Year <= 2019) |>
  count(ISO, Year, Disaster.Type) |>
  summarise(`Country-year-type combinations` = n(),
            `Of those, with more than one event` = sum(n > 1),
            `Largest number of events in one country-year` = max(n)) |>
  pivot_longer(everything(), names_to = "Check", values_to = "Count") |>
  kable(caption = "Repeated disasters in the same country and year")
```

Table 8: Repeated disasters in the same country and year

Check	Count
Country-year-type combinations	647
Of those, with more than one event	115
Largest number of events in one country-year	11

Both builds collapse these to a single indicator rather than a count, so a country-year with three earthquakes is coded 1. The week 3 version does this with `distinct()` before `pivot_wider()`, the agent version with `max()` inside `aggregate()`, and the two give the same answer.

5.3 The conflict threshold

```r
conflict_raw <- read.csv(here("data", "raw", "conflict.csv"))

conflict_cy <- conflict_raw |>
  group_by(iso, year) |>
  summarise(deaths = sum(best, na.rm = TRUE), .groups = "drop")

conflict_cy |>
  summarise(`Country-years in the conflict file` = n(),
            `Reaching 25 deaths` = sum(deaths >= 25),
            `Below 25 deaths` = sum(deaths < 25)) |>
  pivot_longer(everything(), names_to = "Check", values_to = "Count") |>
  kable(caption = "The conflict file collapsed to country-years")
```

Table 9: The conflict file collapsed to country-years

Check	Count
Country-years in the conflict file	3720
Reaching 25 deaths	704
Below 25 deaths	3016

Both builds sum `best` over every conflict active in a country-year and code the exposure as 1 when the total reaches 25 battle related deaths, which is the UCDP definition the paper uses. Summing before thresholding matters: a country with three separate conflicts of ten deaths each is exposed under this rule and would not be if each conflict were tested on its own.

6 One number that does not reconcile with the paper

The paper reports 522 country-years of minor conflict and 148 of war, which is 670 exposed country-years. This data set has 665.

```r
exposed <- conflict_cy |> filter(deaths >= 25)

tibble(
  Source = c("Paper (522 minor + 148 war)",
             "Conflict file, 25 death threshold, all years",
             "Conflict file, 25 death threshold, 2000 to 2019",
             "In final_data.csv after the merge"),
  `Country-years` = c(670,
```

```
                      nrow(exposed),
                      sum(exposed$year >= 2000 & exposed$year <= 2019),
                      sum(own$armedconflict))
) |>
  kable(caption = "Reconciling the exposure count against the paper")
```

Table 10: Reconciling the exposure count against the paper

Source	Country-years
Paper (522 minor + 148 war)	670
Conflict file, 25 death threshold, all years	704
Conflict file, 25 death threshold, 2000 to 2019	665
In final_data.csv after the merge	665

```
exposed |>
  filter(year >= 2000, year <= 2019) |>
  anti_join(own, by = c("iso", "year")) |>
  nrow()
```

```
[1] 0
```

The merge loses nothing: every exposed country-year in 2000 to 2019 survives into `final_data.csv`, and the drop from 704 to 665 is entirely years outside the study window. So the five country-year gap against the paper is not a merge problem and not something either implementation introduced.

The disaster indicators do lose a few country-years at the merge, because some ISO codes in EM-DAT have no World Bank mortality data.

```
tibble(
  Indicator = c("Earthquake", "Drought"),
  `Before the merge` = c(sum(by_year$earthquake), sum(by_year$drought)),
  `In final_data.csv` = c(sum(own$earthquake), sum(own$drought))
) |>
  kable(caption = "Disaster country-years lost to countries with no mortality data")
```

Table 11: Disaster country-years lost to countries with no mortality data

Indicator	Before the merge	In final_data.csv
Earthquake	320	310
Drought	327	325

Those countries have no outcome to contribute and cannot enter the analysis, so dropping them is correct rather than a loss.

7 Conclusion

Two independent implementations, one in the tidyverse and one in base R, produce identical output: 3720 rows, 18 columns, no disagreeing cells. The derived variables behave as intended, the lag resets at each country, and the indicator zeros are true zeros rather than unknowns. No edit to `final_data.csv` is needed.

Two things are worth raising in class. The first is that `Year` and `Start Year` in the EM-DAT file disagree on 25 drought rows, and choosing one over the other moves 21 country-years of the drought indicator. The second is the five country-year gap between this data set's 665 exposed country-years and the paper's 670; the merge accounts for none of it, so it is most likely a difference in the vintage of the UCDP release or in how the countries the paper excludes were handled, and the code cannot settle it.